Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 110404

Subject: Mathematics-III (Probability and Statistics)

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) If the occurrence of one event does not affect the occurrence of the other event, the events are called
(i) Dependent
(ii) Mutually Exclusive
(iii) Independent
(iv) Bayesian

(b) For a binomial distribution
(i) Mean = Variance
(ii) Mean < Variance
(iii) Mean > Variance
(iv) Changes with parameters

(c) The moment generating function of a gamma distribution is $(1-2t)^{-3}$ Then its mean is
(i) 12
(ii) $\frac{1}{12}$
(iii) 6
(iv) $\frac{1}{6}$

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(d) For a standard normal distribution
(i) $\mu = 0$
(ii) $\sigma = 1$
(iii) $\mu = 0$ or $\sigma = 1$
(iv) $\mu = 0$ and $\sigma = 1$

(e) The sum of n independent exponential random variables follow
(i) Exponential distribution
(ii) Gamma distribution
(iii) Normal distribution
(iv) Uniform distribution

(f) For a perfectly symmetrical distribution, the coefficient of skewness is
(i) positive
(ii) negative
(iii) 0
(iv) 1

(g) For a Binomial distribution $B(n, p)$. If mean = 6 and variance = 2, then n and p are
(i) $n = 9,\ p = \frac{2}{3}$
(ii) $n = 12,\ p = \frac{1}{3}$
(iii) $n = 8,\ p = \frac{3}{4}$
(iv) $n = 6,\ p = 1$

(h) The point where the null hypothesis gets rejected is called
(i) Significant value
(ii) Rejection value
(iii) Acceptance value
(iv) Critical value

(i) In fitting a straight line y = ax+ b, what is the formula to find the sum of the Squares of the residuals?
(i) $E = \sum y^2 - a\sum xy - b\sum y$
(ii) $E = \sum y^2 + a\sum xy - b\sum y$
(iii) $E = \sum y^2 + a\sum xy + b\sum y$
(iv) $E = \sum y^2 - a\sum x + b\sum y$

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(j) In a chi-square test of independence, the test statistic based on a contingency table with 5 rows and 4 columns has how many degrees of freedom?

(i) 9 (iii) 12

(ii) 15 (iv) 7

Q.2 (a) A random variable X has the following probability function: [7]

Values of X, x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) Find k and variance of random variable X

(ii) If $P(X \le a) > \frac{1}{2}$, find the minimum value of a

(b) The mean and the variance of a binomial distribution with parameters n and p are 12 and 3. Find the following probabilities: [7]

(i) $P(X<4)$ (ii) $P(4 \le X \le 11)$ (iii) $P(X \ge 7)$.

Q.3 (a) A random variable X is distributed with mean 16 and standard deviation 3. Using Chebyshev's inequality, find lower or upper bounds for the following probabilities: [8]

(i) $P(|X-16| \le 6)$ (ii) $P(|X-16| \le 7.5)$

(b) Find the mean of normal distribution. [6]

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Q.4 (a) Suppose that a random variable X is distributed as exponential distribution with parameter λ. Show that $P(X \le 10) = P(X \le 17 | X > 7)$. [7]

(b) Find the moment generating function of the gamma distribution $f(x) = \frac{1}{\lfloor 1/4} e^{-x} x^{-3/4}$, $x \ge 0$ at the origin. [7]

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Q.5 (a) Let X_1 and X_2 be two independent random variables, both having exponential distributions with parameters λ_1 and λ_2, respectively, and let $Y = X_1 + X_2$. Determine the distribution of random variable Y. [7]

(b) Show that if X_1 and X_2 are standard normal variates with correlation coefficient ρ between them, the correlation coefficient between X_1^2 and X_2^2 is given by ρ^2. [7]

Q.6 (a) The first four moments of a distribution about $x = 2$ are 1, 2.5, 5.5 and 16. Calculate the first four moments about the mean and about origin. [7]

(b) The following table represents the height of a batch of 100 students. Calculate the kurtosis. [7]

Height (in cm)	59	61	63	65	67	69	71	73	75
No. of students	0	2	6	20	40	20	8	2	2

Q.7 (a) If $z = ax + by$ and r is the correlation co-efficient between x and y, then show that $\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2abr\sigma_x\sigma_y$. [7]

(b) Using the method of least squares, derive the normal equations to fit the curve $y = ax^2 + bx$. Hence fit this curve to the following data: [7]

x	1	2	3	4	5	6	7	8
y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Q.8 (a) In a sample of 1,000 people in Maharashtra, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this State at 1% level of significance? [7]

(b) Find mean and variance of Poisson distribution. [7]

Q.9 When the first proof of 392 pages of a book of 1200 pages were read, the distribution of printing mistakes was found to be as follows: [14]

No. of mistakes in a page (x)	0	1	2	3	4	5	6
No. of pages (f)	275	72	30	7	5	2	1

Fit a Poisson distribution to the above data and test the goodness of fit.

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